Beyond 100% Accuracy: Leakage-Aware Evaluation of Engineered, Deep and Foundation-Model Features for Leukemia Detection in Blood Smears

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***Abstract*—Deep-learning models for leukemia detection in blood smears often report 97–100% accuracy, yet few reach clinical use. We test the hypothesis that much of this gap is created by evaluation shortcuts: splitting cells rather than patients, copying a patient's diagnosis onto every cell, and testing within one cohort. Using the public APL-AML dataset (106 patients, 25,724 single-cell images, Discovery and temporally separate Validation cohorts), we compared 93 engineered morphology features, frozen ImageNet EfficientNet-B0 embeddings and their fusion, with logistic regression, LightGBM and gated attention multiple instance learning, under image-level cross-validation, patient-grouped cross-validation and a once-scored external cohort. When the label was the patient's diagnosis, image-level cross-validation reported AUROC 0.94–0.99 while the same models reached 0.66–0.73 on the external cohort, an inflation of 0.25–0.32; for a cell-level label (blast detection) inflation was only 0.01–0.04. Under honest evaluation, engineered features outperformed ImageNet embeddings for external blast detection (AUROC 0.957 vs. 0.920; paired difference $+0.034$, 95% CI $[0.008, 0.065]$), fusion improved over ImageNet embeddings at cell level ($+0.012$) and patient level ($+0.049$, $[0.007, 0.111]$), and stain/color and chromatin-texture features carried most of the engineered signal. Patient-level APL-vs-AML reached external AUROC 0.778 with attention pooling of engineered features, but calibration did not transfer across cohorts (ECE 0.23–0.38). We recommend patient-level splitting, explicit label-level reporting, external testing and calibration reporting for leukemia imaging studies, and release code and split files.**

***Index Terms*—Acute myeloid leukemia, acute promyelocytic leukemia, blood smear, data leakage, deep learning, domain shift, feature engineering, multiple instance learning, model calibration, reproducibility.**

I. Introduction

MICROSCOPIC review of peripheral blood and bone marrow smears is the first step in the diagnosis of leukemia [1]. It is slow, depends on scarce expertise and varies between observers, which has motivated a large body of work on automated, deep-learning-based analysis. Many recent studies report 97–100% accuracy for leukemia detection from blood-smear images [2]–[4], yet such systems rarely reach clinical use.

We argue that a large part of this gap comes from how models are evaluated rather than how they are built. Three practices are common. First, single-cell datasets are split into training and test sets at the image level, so cells of the same patient appear on both sides. Second, a patient-level label such as the diagnosis is copied onto every cell of that patient, so a model can score well by recognizing the patient—stain, slide or acquisition date—instead of the disease. Third, models are tested on held-out images from the same source instead of a separate cohort. These are instances of the leakage and shortcut problems documented across machine-learning-based science [5] and medical imaging [6], [7].

The consequences are visible in recent work. A subject-disjoint re-evaluation of the widely used C-NMC 2019 dataset [8] reached AUROC 0.913 instead of near-perfect values [9], and a DinoBloom-based patient-level model trained on 6,610 individuals from a diagnostic laboratory reached 0.72 accuracy over eight disease classes [10]. What is still missing is a controlled measurement of how much each shortcut inflates reported performance on real patient data, and of which representations hold up once the shortcuts are removed.

This paper makes the following contributions:

1) A controlled leakage audit: on identical data, features and classifiers we compare image-level cross-validation (CV), patient-grouped CV and an untouched external cohort, for a label that is a patient property (diagnosis) and a label that is a cell property (cell type).
2) A comparison of 93 interpretable engineered morphology features, frozen ImageNet CNN embeddings and their fusion under honest evaluation, with feature-group ablations that show where performance comes from.
3) Patient-level APL-vs-AML diagnosis with gated attention multiple instance learning (MIL) [11], evaluated on an external cohort with calibration.
4) An attention audit against technologist cell labels, and a release of code, split files and label maps for comparable future reporting.

II. Related Work

A. Deep Learning on Blood Smears

Most published systems classify single segmented cells with ImageNet-pretrained CNNs, CNN–transformer hybrids or ensembles trained on small public datasets such as ALL-IDB [12], C-NMC 2019 [8] and the ALL image dataset. Reported accuracies cluster between 95% and 100% [2], [3], and hybrid CNN-feature pipelines reach about 88% in three-class settings [4]. The split unit (image or patient) is often not reported.

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TABLE I
APL-AML DATASET COMPOSITION BY COHORT AND DIAGNOSIS.

Cohort	Diagnosis	Patients	Cells	Signed cells	Cells/patient median (range)
Discovery	AML	60	11,523	8,077	134 (98–984)
	APL	22	8,764	4,082	290 (112–1,066)
Validation	AML	12	3,589	1,893	252 (116–554)
	APL	12	1,848	1,580	140 (83–268)
Total		106	25,724	15,632	152 (83–1,066)

B. Patient-Level and Morphology-to-Genotype Models

Attention-based MIL [11] has been used to identify genetic AML subtypes from blood-smear cells [13], APL from smears [14], NPM1 and FLT3-ITD status from marrow whole-slide images [15], [16], and eight hematological malignancy classes in a diagnostic-laboratory cohort [10]. Expert-annotated single-cell datasets underpin these advances [17], [18]. These studies split by patient, but most train and test within one laboratory.

C. Hematology Foundation Models

DinoBloom, a DINOv2 [19] model trained on more than 380,000 cells from 13 public datasets, outperforms generic encoders for cell typing and AML subtyping [20]. Subsequent work adapts DINOv3 with low-rank adaptation and a hierarchical, risk-aware loss [21], and compares DinoBloom, BiomedCLIP and CLIP with retrieval augmentation under a held-out-dataset protocol [22]. LeukemiaAttri records the same films with different microscopes, cameras and magnifications to expose acquisition shift [23].

D. Research Gap

To our knowledge, no study has (i) measured how leakage depends on whether the label belongs to the patient or the cell, (ii) compared engineered and learned representations under identical patient-grouped splits with an external cohort, and (iii) reported calibration and an attention audit for patient-level leukemia classification on the same data.

III. MATERIALS

A. Dataset

We used the public APL-AML dataset of Sidhom *et al.* [14]. It contains single-cell images from Wright-stained peripheral blood smears of 106 adults with acute myeloid leukemia seen at Johns Hopkins Hospital (2010–2020), acquired with a CellaVision DM100 system; 34 patients have acute promyelocytic leukemia (APL) and 72 other AML subtypes. The source defines a Discovery cohort (82 patients, presenting before January 2019) and a temporally separate Validation cohort (24 patients, from January 2019). Cells are stored as JPEG files of about 360×363 pixels. 15,632 cells carry a cell type verified by the CellaVision operator ("signed"); 10,092 are unsigned. Table I and Fig. 1 summarize the composition.

TABLE II
LEARNING TASKS. CELL COUNTS ARE DISCOVERY / VALIDATION.

Task	Unit	Positive class	Discovery / Validation
T1 Diagnosis-on-cells	Cell	Cell from APL patient (43.2%)	20,287 / 5,437
T2 Blast detection	Cell	Signed blast (28.0%)	9,694 / 2,863
T3 Patient diagnosis	Patient	APL (22 of 82)	82 / 24 patients

B. Data Limitations and Challenges

- *Class imbalance.* APL makes up 27% of Discovery patients but 50% of Validation patients. At cell level, 43.2% of Discovery cells come from APL patients (T1) and 28.0% of signed Discovery cells are blasts (T2). Rare cell types (basophils, plasma cells, promonocytes) have fewer than 75 examples.
- *Bag-size confounding.* Discovery APL patients have about twice as many cells as AML patients (median 290 vs. 134), while the opposite holds in Validation (140 vs. 252). Any model sensitive to the number of cells would learn a pattern that reverses at test time; our aggregators are invariant to bag size.
- *Acquisition batch effect.* The original authors report that all pre-2018 Discovery smears were APL [14], so acquisition period is partly confounded with diagnosis. This favors shortcut learning whenever patients leak across splits.
- *Cell-type composition differs by diagnosis.* Signed monocytes are almost absent in APL (Fig. 1b), so a cell-type mixture is itself a diagnostic signal.
- *Label provenance.* Cell types are CellaVision pre-classifications verified by an operator, not consensus labels; 25 artefacts and 31 unidentified cells are present.
- *Image quality.* Ten files are truncated and unreadable, and 54 images are full fields (about 2,900 pixels) rather than single-cell crops.
- *Single institution, small external cohort.* Validation is temporal, not multi-site, and contains 24 patients, which limits the precision of patient-level estimates.

C. Tasks

Three tasks were defined on the same data (Table II). T1 reproduces the common practice of labelling cells with the patient's diagnosis. T2 uses a label that is a property of the cell itself; artefacts, unidentified cells, smudge cells and platelet categories were excluded. T3 is the clinical task.

D. Working Environment

All experiments ran on a CPU-only Linux workspace (Table III); no GPU was used. Feature extraction is the dominant cost; once features are cached, every cell-level configuration trains and evaluates in under one minute and every patient-level configuration in about one minute.

IV. METHODS

Fig. 2 shows the preprocessing pipeline, Fig. 4 the model architecture and Fig. 5 the evaluation protocol.

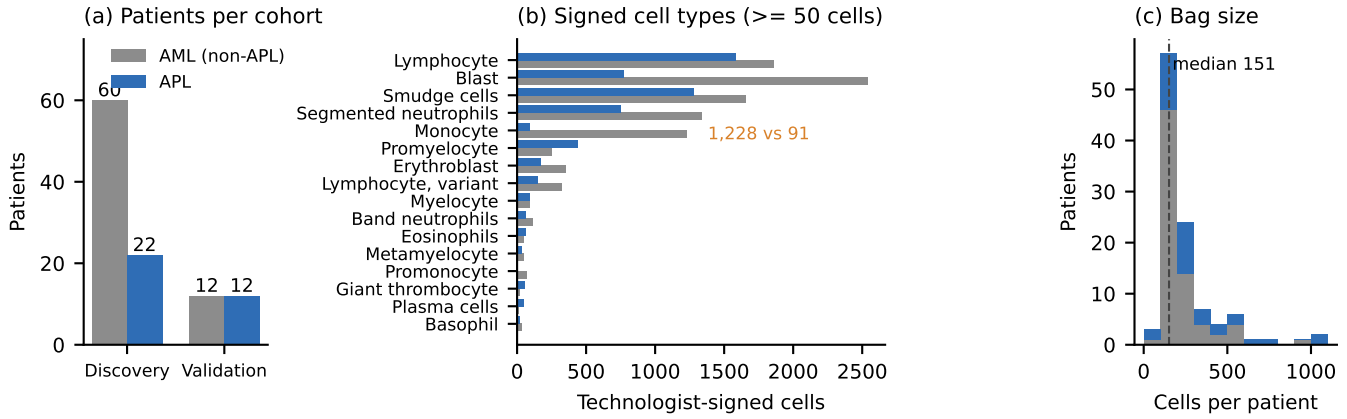


Fig. 1. Dataset composition. (a) Patients per cohort and diagnosis. (b) Technologist-signed cell types with at least 50 cells, by diagnosis; monocytes are strongly depleted in APL patients (1,228 vs. 91 cells). (c) Number of cells per patient (bag size).

TABLE III
WORKING ENVIRONMENT AND COMPUTATIONAL COST.

Item	Specification
Processor	Intel Xeon @ 2.80 GHz, 2 vCPUs, no GPU
Memory	7.8 GB RAM
Operating system	Ubuntu 24.04.5 LTS, Linux kernel 6.18
Language	Python 3.13.16
Libraries	NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, scikit-learn 1.9.1 [24], scikit-image 0.26.0, LightGBM 4.7.0 [25], PyTorch 2.14.1 (CPU), torchvision 0.29.1, timm 1.0.30, Pillow 12.3.0
Random seeds	0, 1, 2
Engineered features	16.1 min for 25,724 cells (2 processes)
EfficientNet-B0 embeddings	51.2 min for 25,724 cells (2 threads)
Cell-level model (5-fold CV)	11–40 s (LR), 15–1,000 s (LightGBM)*
Attention MIL (5-fold CV)	33–61 s per seed

*Upper values were measured while other jobs shared the CPU.

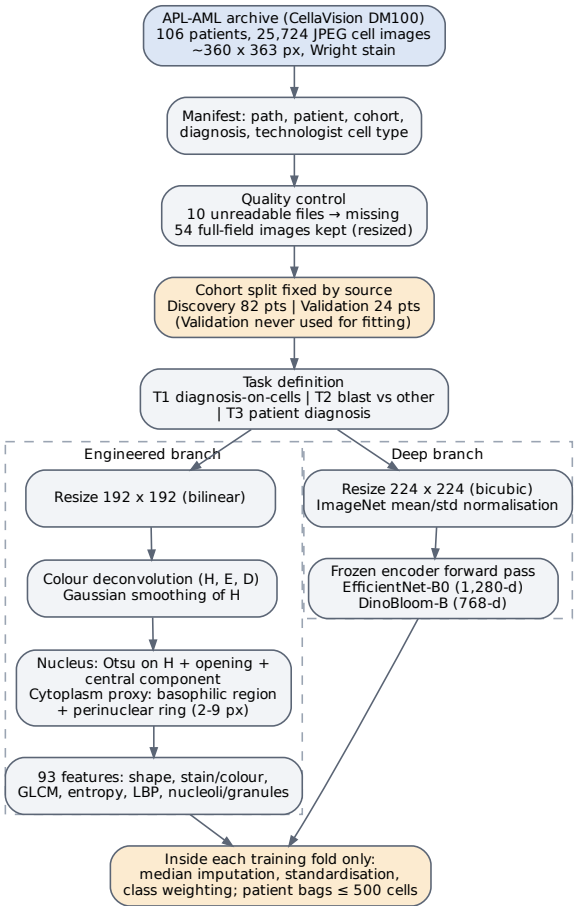


Fig. 2. Data preprocessing pipeline. Orange boxes mark the two steps that prevent leakage: the fixed Discovery/Validation split and fitting every statistic inside the training fold only.

A. Preprocessing

A manifest records each image's path, patient, cohort, diagnosis and signed cell type. Unreadable files are kept as missing rows and imputed inside each training fold; full-field images are resized like all others. No image is discarded on the basis of its label, and the Validation cohort is never used for fitting, threshold selection or calibration.

B. Engineered Features

Images were resized to 192×192 pixels. Color deconvolution [26] separated hematoxylin, eosin and residual channels; the hematoxylin channel was smoothed (Gaussian, $\sigma = 1$) and thresholded with Otsu's method [27], followed by morphological opening and selection of the component nearest the image center. Because pale cytoplasm could not be separated reliably from neighboring red cells by color, we did not claim a full cell outline. Instead, a *basophilic region*—blue-dominant pixels within one nuclear radius of the nucleus—and a *perinuclear ring* 2–9 pixels wide (bright background excluded) served as cytoplasm proxies (Fig. 3). Ninety-three features were computed in six groups:

- *Nuclear shape* (9): area, perimeter, circularity, eccentricity, solidity, extent, axis ratio, major axis and number of nuclear fragments;
- *Basophilic region* (10): the same shape descriptors, nucleus-to-region ratio and rim size;

- *Stain and color* (26): mean and standard deviation of hematoxylin and eosin optical density and of CIE $L^*a^*b^*$ and HSV saturation in nucleus and ring, plus nucleus–ring contrasts;
- *Chromatin texture* (16): gray-level co-occurrence matrix contrast, dissimilarity, homogeneity, energy and correlation [28] at distances 1, 2 and 4 averaged over four angles, and nuclear intensity entropy;
- *Local binary patterns* (28): uniform LBP histograms [29] with $(P, R) = (8, 1)$ and $(16, 2)$;
- *Nucleoli and granules* (4): count and area of pale nuclear spots and dark cytoplasmic specks.

No feature uses image size, file format or dataset identity.

C. Deep Representations

A frozen EfficientNet-B0 [30] pretrained on ImageNet (timm `efficientnet_b0.ra_in1k`) produced a 1,280-dimensional pooled embedding per cell from 224×224 bicubic-resized, ImageNet-normalized input. *Fusion* denotes the column-standardized concatenation of engineered features and embeddings. The pipeline also supports the DinoBloom hematology foundation model [20], loaded as in its official implementation; its results are left to future work because its weights were not reachable from the compute environment.

D. Classifiers

Cell-level models were L2-regularized logistic regression (LR; $C = 0.1$, class-balanced weights) and LightGBM [25] (400 trees, learning rate 0.05, 31 leaves, 0.8 row and column subsampling, class-balanced). Each model is a scikit-learn pipeline with median imputation (and standardization for LR), so all statistics are fitted on training rows only.

For T3, each patient is a bag of at most 500 randomly sampled cells. The *pooled* baseline summarizes each feature by its mean, standard deviation and 10th/50th/90th percentiles and applies LR. The *gated attention MIL* model [11] embeds each cell as $\mathbf{h}_k = \mathrm{ReLU}(\mathbf{W}\mathbf{x}_k)$ (128 units, dropout 0.25), computes

$$a_k = \frac{\exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_k) \odot \sigma(\mathbf{U}\mathbf{h}_k))\}}{\sum_j \exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_j) \odot \sigma(\mathbf{U}\mathbf{h}_j))\}}, \qquad (1)$$

and classifies the bag embedding $\mathbf{z} = \sum_k a_k \mathbf{h}_k$ with a linear layer. It is trained with class-weighted binary cross-entropy, AdamW (learning rate 2×10^{-4}, weight decay 10^{-3}) for 40 epochs with 20% random cell dropout; three seeds are averaged for the external test.

E. Evaluation Protocol

For every model, split type and seed we ran 5-fold CV on the Discovery cohort with either image-level stratified folds or patient-grouped stratified folds (Fig. 5). Under image-level splits all 82 patients appeared in more than one fold; under grouped splits none did. Within each training fold an inner 80/20 split of the same type was used: the model was fitted on 80%, and the remaining 20% selected a calibration temperature [31] and a decision threshold (sensitivity 0.95 for cell tasks, Youden's J for T3). For the external evaluation each model was trained on the whole Discovery cohort with the same inner procedure and applied once to the Validation cohort.

F. Metrics and Statistics

We report AUROC, sensitivity, specificity, expected calibration error (ECE, 15 bins) and Brier score. Confidence intervals (95%) are obtained by bootstrap resampling of patients [32]—1,000 resamples for cell tasks and 2,000 for T3—because cells of one patient are correlated. Differences in AUROC between two models on the same test set are assessed with a two-sided paired patient-level bootstrap. Leakage inflation is defined as image-level CV AUROC minus external AUROC.

V. Results

A. Leakage Audit

Table IV and Fig. 6 compare the three evaluation protocols on identical features and classifiers. For T1, where each cell inherits its patient's diagnosis, image-level CV reported AUROC 0.938–0.987. Patient-grouped CV lowered this to 0.867–0.901, and the external cohort to 0.658–0.734: image-level CV overstated external performance by 0.25–0.32. For T2, where the label is the cell's own type, the three protocols nearly agreed (image-level 0.956–0.978, grouped 0.927–0.954, external 0.920–0.957; inflation 0.013–0.036). Models trained under leaky and honest protocols achieved similar external AUROC (e.g., EfficientNet-B0 on T1: 0.721 vs. 0.722 for seed 0); leakage therefore inflated the reported number rather than changing the model.

B. Engineered Versus Deep Representations

On external blast detection (T2) the best model used engineered features with LightGBM (AUROC 0.957, 95% CI $[0.931, 0.977]$), ahead of frozen EfficientNet-B0 (0.920, $[0.874, 0.956]$) and fusion (0.932). In paired comparisons (seed 0, Fig. 7) engineered features improved on EfficientNet-B0 by $+0.034$ ($[0.008, 0.065]$, $p = 0.002$) and fusion improved on EfficientNet-B0 by $+0.012$ ($[0.003, 0.024]$, $p = 0.004$). On T1 fusion gave the highest external AUROC (0.734 vs. 0.694 for EfficientNet-B0), but the paired difference was not significant ($+0.033$, $[-0.016, 0.072]$).

C. Performance Impact of Feature Groups

Table V and the lower part of Fig. 7 show which engineered groups matter. For T2, removing stain and color features cost -0.034 AUROC ($[-0.060, -0.015]$), chromatin texture -0.012 and the basophilic region -0.005; nuclear shape, local binary patterns and nucleoli/granule proxies were redundant given the other groups. Stain and color features alone reached 0.912. For T1 no single removal changed AUROC significantly and the confidence intervals were wide, consistent with a weak, patient-level signal.

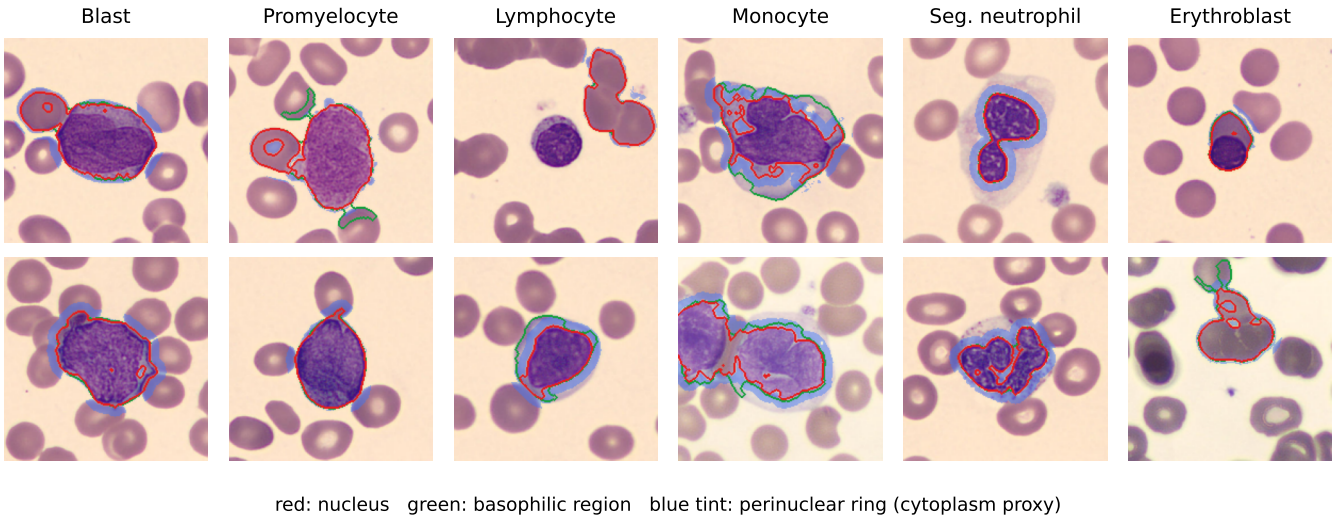


Fig. 3. Segmentation used by the engineered features on example cells (files at the 25th and 75th percentile of each class list). Red: nucleus; green: basophilic region; blue tint: perinuclear ring. Two failures are shown deliberately: the upper lymphocyte and lower erythroblast masks fall on adjacent red cells.

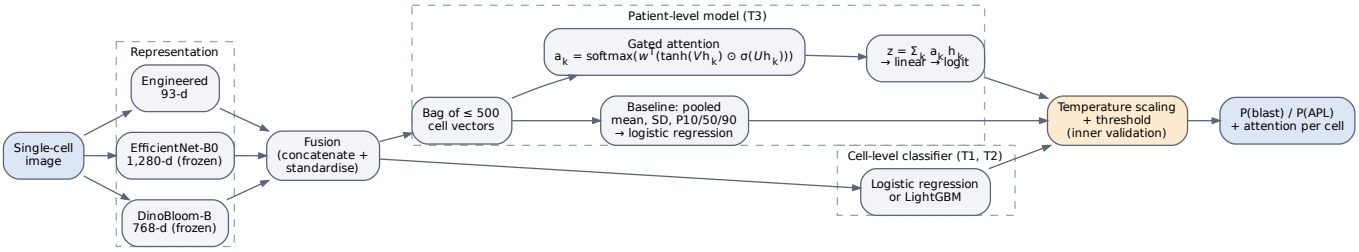


Fig. 4. Model block diagram. Cell images are represented by engineered features, frozen CNN embeddings or both; a cell-level classifier serves T1/T2, and either pooled statistics or gated attention MIL aggregates a patient's cells for T3. Temperature and decision threshold are fitted on an inner validation split of the training data.

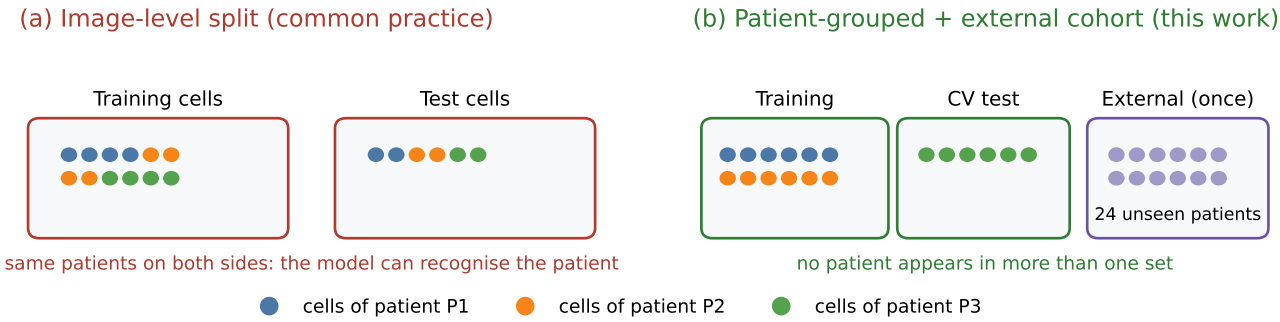


Fig. 5. Evaluation protocols. (a) Image-level splitting, common in published work, places cells of the same patient in training and test sets. (b) Patient-grouped splitting with an external cohort, used here; the external cohort is scored once.

D. Patient-Level Diagnosis

Table VI reports T3. Models with ImageNet features had the highest Discovery CV AUROC (0.885–0.910) but the lowest external AUROC (0.625–0.674), a drop of 0.24–0.27. Attention MIL on engineered features had the lowest CV AUROC (0.827 ± 0.003) but the highest external AUROC (0.778, $[0.566, 0.943]$; sensitivity and specificity 0.75) and the smallest drop (0.049). With 24 external patients this advantage over EfficientNet-B0 MIL was not significant ($+0.153$, $[-0.098, 0.406]$), whereas fusion improved significantly over EfficientNet-B0 within attention MIL ($+0.049$, $[0.007, 0.111]$, $p = 0.032$).

E. Calibration

Table VII shows external calibration before and after temperature scaling fitted on Discovery. Scaling roughly halved the ECE of deep and fused models on T1 (e.g., 0.207 to 0.099 for EfficientNet-B0) but changed T2 by less than 0.01,

TABLE IV
LEAKAGE AUDIT AT CELL LEVEL (AUROC, MEAN OF THREE SEEDS). EXTERNAL = VALIDATION COHORT, MODEL TRAINED WITH PATIENT-GROUPED PROTOCOL; 95% CI FROM PATIENT-LEVEL BOOTSTRAP.

Task	Model	Image-level CV	Grouped CV	External [95% CI]	Sens.	Spec.	ECE	Inflation†
T1 Diagnosis-on-cells	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	0.69	0.55	0.146	+0.318
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	0.73	0.46	0.128	+0.280
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	0.81	0.43	0.081	+0.281
	Fusion + LR	0.985	0.901	**0.734 [0.595, 0.848]**	0.78	0.54	0.102	+0.250
T2 Blast detection	Engineered + LightGBM	0.978	0.954	**0.957 [0.931, 0.977]**	0.99	0.62	0.075	+0.021
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	0.98	0.70	0.110	+0.013
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	0.96	0.64	0.109	+0.036
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	0.96	0.68	0.091	+0.033

†Image-level CV AUROC minus external AUROC.

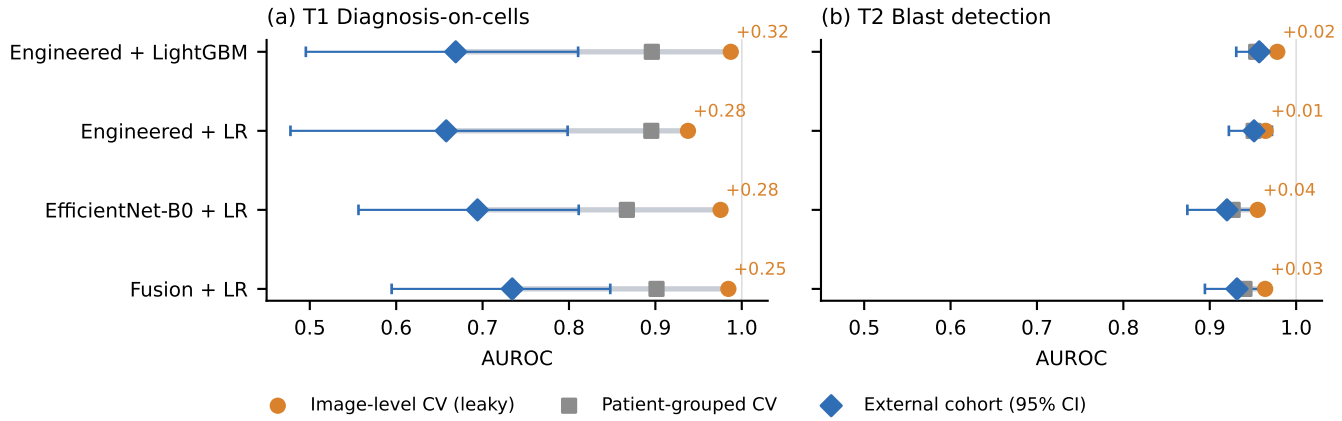


Fig. 6. Leakage audit (AUROC, mean of three seeds). Orange numbers give inflation, i.e., image-level CV minus external AUROC. (a) When the patient's diagnosis is copied onto cells, image-level CV overstates external performance by 0.25–0.32. (b) For a cell-level label the protocols nearly agree.

TABLE V
ENGINEERED FEATURE-GROUP ABLATION ON THE EXTERNAL COHORT (LOGISTIC REGRESSION, EXTERNAL AUROC). Δ: CHANGE VS. ALL 93 FEATURES WITH PAIRED PATIENT-BOOTSTRAP 95% CI.

	T2 Blast detection		T1 Diagnosis-on-cells	
Setting (n features)	AUROC	Δ [95% CI]	AUROC	Δ [95% CI]
All engineered features (93)	0.950	–	0.671	–
Remove one group				
– Nuclear shape (84)	0.949	-0.001 [-0.003, +0.001]	0.681	+0.010 [-0.004, +0.024]
– Basophilic region (83)	0.945	-0.005 [-0.009, -0.003]*	0.665	-0.005 [-0.025, +0.010]
– Stain and colour (67)	0.916	-0.034 [-0.060, -0.015]*	0.548	-0.123 [-0.272, +0.046]
– Chromatin texture (77)	0.939	-0.012 [-0.023, -0.004]*	0.675	+0.004 [-0.007, +0.016]
– Local binary patterns (65)	0.947	-0.004 [-0.010, +0.002]	0.670	-0.001 [-0.019, +0.020]
– Nucleoli and granules (89)	0.951	+0.001 [-0.002, +0.004]	0.671	+0.000 [-0.006, +0.006]
Use one group only				
Nuclear shape (9)	0.810	-0.140 [-0.212, -0.082]*	0.549	-0.121 [-0.283, +0.062]
Basophilic region (10)	0.823	-0.127 [-0.198, -0.069]*	0.579	-0.092 [-0.242, +0.073]
Stain and colour (26)	0.912	-0.038 [-0.078, -0.013]*	0.682	+0.011 [-0.032, +0.041]
Chromatin texture (16)	0.802	-0.148 [-0.221, -0.086]*	0.526	-0.145 [-0.331, +0.067]
Local binary patterns (28)	0.793	-0.157 [-0.225, -0.098]*	0.581	-0.090 [-0.241, +0.081]
Nucleoli and granules (4)	0.860	-0.090 [-0.142, -0.054]*	0.500	-0.171 [-0.365, +0.057]

* $p < 0.05$, paired patient bootstrap.

and residual ECE remained 0.06–0.13 at cell level. At patient level, ECE on the external cohort was 0.23–0.38 for all models (Table VI).

F. Attention Audit

For the best patient-level model (engineered features, attention MIL), we compared attention weights with technologist cell types in external patients. Relative attention (1 = uniform) was highest for cells signed as monocytes (2.8 in AML, 3.8 in APL patients) and below average for signed blasts (0.42–0.45)

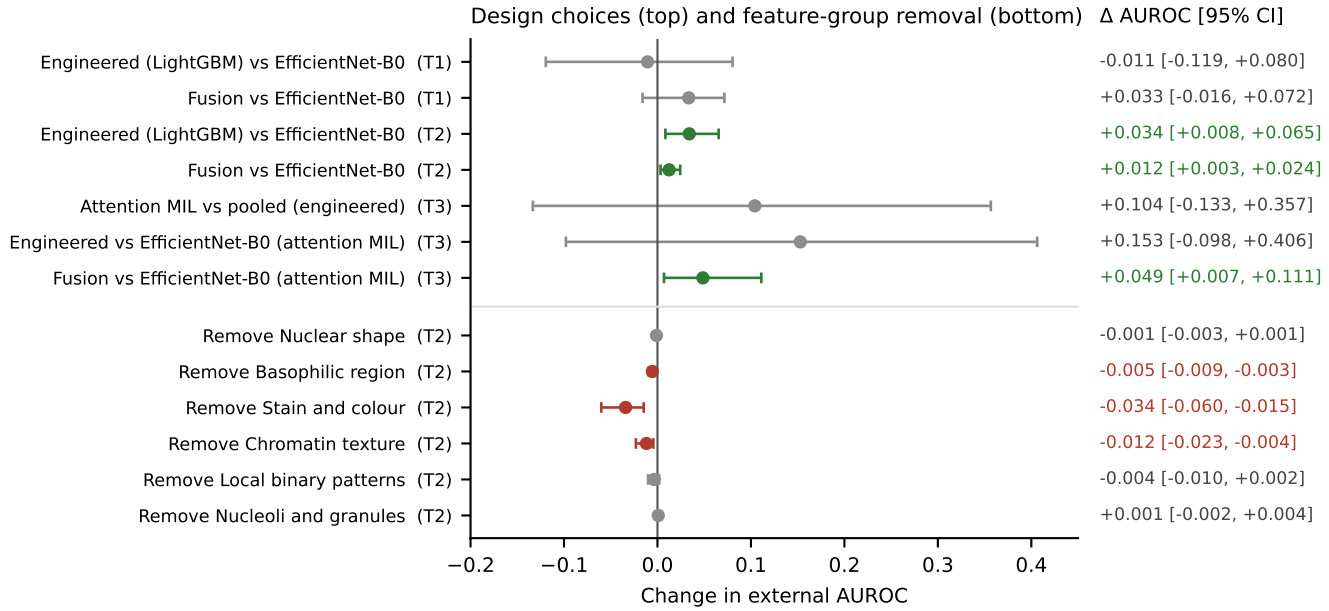


Fig. 7. Performance impact on the external cohort. Top: paired differences between design choices (T1/T2 seed 0; T3 three-seed ensembles). Bottom: removal of each engineered feature group for blast detection. Bars are 95% confidence intervals from a paired patient-level bootstrap.

TABLE VI
PATIENT-LEVEL APL VS AML (T3). CV: DISCOVERY, MEAN ± SD OF THREE SEEDS. EXTERNAL: VALIDATION COHORT (24 PATIENTS), 95% CI BY PATIENT BOOTSTRAP.

Features	Aggregation	CV AUROC	External AUROC [95% CI]	Sens./Spec.	ECE
Engineered	Pooled + LR	0.867±0.059	0.674 [0.440, 0.896]	0.67/0.58	0.380
EfficientNet-B0	Pooled + LR	0.909±0.018	0.639 [0.412, 0.857]	0.75/0.50	0.304
Fusion	Pooled + LR	0.910±0.018	0.667 [0.443, 0.874]	0.50/0.67	0.319
Engineered	Attention MIL	0.827±0.003	**0.778 [0.566, 0.943]**	0.75/0.75	0.300
EfficientNet-B0	Attention MIL	0.885±0.021	0.625 [0.385, 0.844]	0.58/0.58	0.231
Fusion	Attention MIL	0.886±0.039	0.674 [0.437, 0.881]	0.58/0.58	0.274

TABLE VII
EXTERNAL CALIBRATION ERROR (ECE, 15 BINS) BEFORE AND AFTER TEMPERATURE SCALING FITTED ON DISCOVERY.

	T1 Diagnosis-on-cells		T2 Blast detection	
Model	Raw	Scaled	Raw	Scaled
Engineered (LR)	0.130	0.110	0.097	0.096
Engineered (LightGBM)	0.172	0.130	0.052	0.058
EfficientNet-B0 (LR)	0.207	0.099	0.105	0.102
Fusion (LR)	0.203	0.108	0.088	0.082

and promyelocytes (1.08–1.23). Because signed monocytes are almost absent from APL patients in the Discovery cohort (Fig. 1b), the model appears to use the presence of monocyte-like cells as evidence against APL. This is biologically plausible but may also reflect hypogranular APL promyelocytes signed as monocytes; it requires expert review.

VI. COMPARISON WITH PREVIOUS WORK

Table VIII places our results beside published work. The near-perfect accuracies reported on C-NMC and the ALL image dataset [2], [3] were obtained without a stated patient-level split; our audit shows that this setting can inflate AUROC by up to 0.32 when the label is a patient property. Where evaluation is comparable, our results are consistent with honest prior estimates: subject-disjoint testing on C-NMC gave AUROC 0.913 [9], and the original APL-AML study reported a cell-level AUC that fell from 0.890 in Discovery to 0.743 in Validation [14].

The improvements of this work are therefore not a higher headline accuracy but (i) a quantified, label-dependent estimate of leakage inflation on real patient data; (ii) an external cell-level AUROC on the APL-AML Validation cohort of 0.734 for fused frozen features, at the level of the 0.743 reported for an end-to-end trained CNN [14], without training a deep network; (iii) a significant external gain of engineered over ImageNet features for blast detection (+0.034) and of fusion over ImageNet features at cell (+0.012) and patient level (+0.049); (iv) feature-group ablations showing that stain/color and chromatin texture carry the signal; and (v) calibration and attention analyses that prior work on this dataset did not report.

TABLE VIII
COMPARISON WITH PREVIOUS WORK. METRICS ARE AS REPORTED BY EACH STUDY; THEY ARE NOT DIRECTLY COMPARABLE ACROSS DATASETS.

Study	Data	Evaluation	Reported result	Limitation addressed here
Jammal & Dahab, 2025 [2]	C-NMC (cells)	Split unit not stated	100% accuracy	Patient leakage not excluded; no external test
Jammal *et al.*, 2025 [3]	ALL image / C-NMC	5-fold CV, split unit not stated	Accuracy 1.000 / 0.972	Same as above
Kasim *et al.*, 2025 [4]	ALL-IDB2 + AML-LMU, 390 images	Image-level, two sources	88% (3 classes)	Class confounded with source
Albzour, 2026 [9]	C-NMC	Subject-disjoint, unseen subjects	AUROC 0.913	Single label type; no patient-level model
Sidhom *et al.*, 2021 [14]	APL-AML (same data)	Discovery CV, temporal validation	Cell AUC 0.890 $\rightarrow$ 0.743	Leakage not quantified; no calibration
Dasdelen *et al.*, 2026 [10]	MLL, 6,610 individuals; external APL-AML	Patient-level, two external sets	8-class acc. 0.72; APL-AML acute-leukemia sens. 0.76	Not designed to measure leakage
This work	APL-AML (106 patients)	Image vs. grouped CV vs. external; patient bootstrap	External AUROC: T1 0.734, T2 0.957, T3 0.778; inflation 0.25–0.32 (T1)	–

VII. DISCUSSION

A. Why Leakage Depends on the Label

Leakage inflated performance by 0.25–0.32 AUROC when cells were labelled with their patient's diagnosis and by at most 0.04 when the label was the cell's own type. When a label belongs to the patient, every cell carries patient-specific signals—stain intensity, smear preparation, acquisition period—that predict the label perfectly within a leaky split. The APL-AML Discovery cohort makes this especially easy because acquisition period is partly confounded with diagnosis [14]. When the label belongs to the cell, these signals are shared by positive and negative cells of the same patient and offer no shortcut. Reported single-cell accuracies should therefore be read together with the label level and the split unit.

B. Why Engineered Features Transfer

ImageNet embeddings achieved the best internal scores but generalized worst at patient level, and engineered features significantly outperformed them for external blast detection. Engineered features encode a small number of stain-normalized, clinically meaningful quantities (optical density, chromatin texture, nuclear and basophilic-region geometry), whereas 1,280-dimensional generic embeddings can absorb cohort-specific appearance. The ablations show that stain/color and chromatin texture carry most of the transferable signal, in line with how hematologists distinguish blasts. Fusion was the most robust choice overall, improving significantly over ImageNet embeddings at cell and patient level.

C. Calibration

Temperature scaling fitted on Discovery did not make probabilities reliable on the Validation cohort, especially at patient level. Calibration should therefore be re-assessed, and if necessary re-fitted, at each deployment site before probabilities are used for triage.

D. Limitations

This study uses one institution with a temporally separate, 24-patient external cohort, so patient-level confidence intervals are wide. The hematology foundation model DinoBloom was implemented but not evaluated because its weights were not reachable from the compute environment; the cross-dataset experiments with C-NMC, AML-Cytomorphology_LMU, BMC and AML-Hehr are part of the released protocol but not of these results. Paired tests at cell level use one seed. Engineered cytoplasm features rely on proxies rather than a full cell outline, and segmentation fails on some small cells (Fig. 3). Cell-type labels are operator-verified CellaVision classifications, not consensus labels.

VIII. CONCLUSION

Reported accuracy for leukemia detection from blood smears depends strongly on the evaluation protocol. When a patient's diagnosis is copied onto cells and cells are split at random, cross-validation overstates external AUROC by about 0.3; for cell-level labels the overstatement is small. Under honest evaluation, interpretable engineered features match or exceed frozen ImageNet embeddings, fusion is the most robust representation, and patient-level attention MIL on engineered features generalizes best to a later cohort, although calibration does not transfer. We recommend that leukemia imaging studies split by patient, state whether labels are cell- or patient-level, test on a separate cohort or site, and report calibration with patient-level confidence intervals. Future work will add hematology foundation-model features and multi-site external validation.

DATA AND CODE AVAILABILITY

The APL-AML dataset is available at https://github.com/sidhomj/DeepAPL. Code, split files, label maps, per-model summaries and per-cell attention weights (LeukBench) are released at [repository URL]. The study is reported following TRIPOD+AI [33].

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